

MAJOR subdivisions of this paper are concerned with: (1) the place of linguistics in cultural anthropology; (2) trends in linguistics. Neither parts are to be taken in an excluding sense. Besides cultural anthropology under 1, we include also reference to archaeology; besides trends under 2, we include reference to continuities, as the undiminished flow of work based on the comparative method in Indo-European.

Discussion in this paper is placed under italicized statements of a general nature. These general points or general assertions were first assembled by the writers and then discussed between them; rather to their surprise, some points which appeared promising for discussion proved quite sterile when discussion was actually attempted. This is reflected in our paper by the brevity of some of the discussion paragraphs which do no more than list or outline agenda. Other discussion paragraphs are a summary of the more fruitful considerations that preceded this writing.

1. LINGUISTICS AND CULTURAL ANTHROPOLOGY

The data of linguistics and of cultural anthropology are largely the same.

Human behavior, as well as (or rather, which includes) behavior between humans, is never purely verbal; nor, in the general case, is it non-verbal. Linguists characteristically study only that part of a situation which we here call verbal. Cultural anthropologists often segregate the non-verbal from the verbal, relegating the latter to special chapters or volumes (such as folklore), as contrasted with chapters devoted to various aspects of material culture, such as house types; one might infer from some ethnographies that houses are built in sullen silence.

There are exceptions: these we call ethno-linguistic because they attempt to integrate the verbal and non-verbal aspects of behavior, whether in a single historical problem or in a single synchronic situation. We are more indebted to ethnographers than to linguists for contributions which have been made thus far to ethno-linguistics. The main contributions have been historical.

For native America, Sapir summarized the work of a generation of scholars in his "Time Perspective,"¹ in general showing interrelations of languages, cultures, and populations. For reliability in relative chronology, Sapir rates inferences based on linguistic data second only to those based on archaeological data (though admitting that reliability is enhanced when linguistic and archaeological work are correlated). In dealing with the less remote history of Indians of California, Kroeber² controls both cultural and linguistic data.

Much good interpretation has been made in reconstructing early aboriginal

¹ Sapir, 1916.

² Kroeber, 1925.

history in the New World; nevertheless, the more successful use of historical ethno-linguistics has been in the Old World. Migrations have been traced by means of place names, as for the Phoenicians in the Mediterranean Basin.

Analysis of proper names has been used as evidence for the ethnic composition of populations, as, for example, the evidence for Syria and Palestine in the El-Amarna tablets written in the middle of the second millennium B.C. Thousands of names occur in these tablets; many are Semitic names of various branches, some are Hittite, some are Hurrian. The information thus derived concerning the distribution and relative numbers of Semites, Hittites, Hurrians, and others is in part corroborated by archaeological and other historical data. The most famous use of historical ethno-linguistics is in the investigation of the last common home of the Indo-Europeans, and in the use of cognate terms from the various Indo-European branches to give information on the culture then obtaining.

The linguistic parts of such historical reconstructions call for great exactness. And if either sobriety in cultural interpretation, or sophistication in comparative linguistics is lacking, the results are inevitably absurd, as for example, in the Japhetic theory of the Marr Institute.

Compared to the very specific historical work of which examples are given above, the work in non-historical, that is to say, synchronic ethno-linguistics has so far been either particularistic or programmatic. Thus, particular parts of full-length ethnographies actually integrate the non-verbal and verbal aspects of culture, namely the parts devoted to ceremonies and to some aspects of social organization, such as kinship systems; the Dakota materials prepared by Ella Deloria (now in manuscript in the Boas Collection of the American Philosophical Society) go beyond these particular parts of the culture and attempt to give, for all parts of the culture treated, commentary from persons in the culture, and in a situational context.

Malinowski, in his last monographic work,³ exhibits a programmatic interest in ethno-linguistics which is on the whole good and in part brilliant; he also makes a futile attempt to integrate non-verbal and verbal aspects of Trobriand Island culture: futile, perhaps, because the integration is attempted without knowledge of Trobriand linguistic structure.⁴

³ Malinowski, 1935.

⁴ For the sake of brevity and comparability with our recent paper in the *Southwestern Journal of Anthropology* (Vol. 1, pp. 455-465, 1945), we paraphrase closely Malinowski's methodological points from his "Coral Gardens and Their Magic" (New York, 1935); our paraphrase of Malinowski is in italics, with parenthetical page references to volume two of "Coral Gardens"; our comment is in roman.

Language must be linked up with all other aspects of human culture (vii). To make the correlation between L and X, Y, Z, requires structural statements of L and of X, Y, Z, as well as a technique of correlation.

The drift of most modern missionary writings on linguistics is towards the method of cultural

The techniques of linguistics and of cultural anthropology are in general different.

Linguistic techniques enable a worker to state the parts of the whole (for any one language), and to give the distribution of the parts within the whole. This provides criteria of relevance: it is possible to distinguish sharply between

interpretation of language which is also adopted in "Coral Gardens and Their Magic" (ix). The missionary, the administrator, the trader and even the traveller needs to translate the white man's point of view to the native, while the anthropologist needs to translate the native point of view to the white man (x). It is assumed by Malinowski that this problem of translation may be met by extensive commentary which combines both linguistic information and cultural information.

The only correct presentation of linguistic material is in combination with ethnographic description: such a presentation follows the technique of the field-worker, seeing and hearing at the same time and recording in full without trying to be concise; material illuminated from two sides will stand out, so to speak, stereoscopically (3). Since ethnographic material is allowed separate presentation by Malinowski, it would seem somewhat partisan of him to insist on a dependent status for linguistic material; but the fact remains that activities observed in the field are often ethno-linguistic situations, and that the final presentation of life as it is lived loses color because scholars select data according to their specialization, reporting either the ethnographic or the linguistic, but not both.

Language is the ethnographer's most important tool: through his practical handling of native grammar and vocabulary he can ask clear questions and receive relevant answers (4). This is a practical question rather than an ethno-linguistic question.

The methodological classification of utterances is three-fold: statement in answer to a direct question (definition texts); volunteered statement; traditional sayings (magical formulae, gardening cries and ditties) (5). The first class of utterance constituting definition texts comprises three-fourths of all texts listed by Malinowski; most of his materials are therefore scarcely ethno-linguistic because in the definition text the Trobriand Islander does not exhibit verbal behavior in correlation with a non-verbal expression of culture. Instead, he uses his own language to explain to someone outside of his culture, namely Malinowski, how or why that culture is as it is. This would be taken for granted by anyone living inside the culture,—that is to say, the definition text must be a Trobriand Island innovation occasioned by the arrival and demands of Malinowski. The second class of utterance, volunteered statement, is usually but not always marked as belonging to this class. The third class of utterance, traditional sayings, would usually be classified as folklore. And folklore may enter into an ethno-linguistic situation only when the things said are correlated with non-verbal cultural behavior. In so far as Malinowski has found and noted such correlations, he has succeeded in presenting ethno-linguistic situations. In folklore, then, a rather conventional field in anthropology, Malinowski is most fertile.

Language is more than a means to an end: it stands in a definite relation to the life of the people who speak it; thus, when a magician declares, "I cut thee—my garden site; I make thy belly blossom with my charmed axe . . ." he is definitely commenting on his actions (6). Malinowski is not content with a simple analysis into two factors, verbal and non-verbal behavior; he warns against the danger of imagining "that language is a process running parallel and exactly corresponding to mental process" (7), and in order to avoid "the danger" he sets up a third factor, "the mental reality of man."

Language is an adjunct to bodily activities, an indispensable ingredient of all concerted human action (7); it does not always serve to communicate, but is always part of concerted activity (8). To exemplify this distinction Malinowski cites an ethno-linguistic situation in which some old men go off to survey an area for agricultural purposes; in this, Malinowski equates technical terminology with gestures and other kinds of non-verbal behavior such as blazing trees and cutting saplings.

what is and what is not linguistic. Such criteria are lacking in ethnographies where culture traits are none too clearly distinguished from culture complexes and where a given segment of behavior may be regarded by one worker as an expression of culture, by another as an expression of personality; another segment of behavior, thought to be entirely physiological (as morning sickness in

But communication is not lacking here; rather everything said is in the context of the survey, so that if the words were recorded alone, out of their context, they would appear as gibberish.

Speech interwoven with manual behavior is primarily used for the achievement of a practical result; secondarily, it also fulfills an educational purpose in that the older and better-informed men hand on the results of their past experiences to the younger ones (8). The diagnostic feature of an ethno-linguistic situation is for Malinowski some teleology, some purpose, some function. Whether a boundary is surveyed, a garden area cleared, a fence or arbor constructed, such work for one objective "would be impossible without speech." This is based on the practice of one culture; we have observed members of another culture, the Ojibwa north of the Great Lakes, perform parallel tasks silently. A group of people can work together without chattering.

Multiplicity of meanings will be found a characteristic of most native words, but in no case is there any confusion in the mind of the speaker as to which of several distinct realities he wishes to indicate by the use of one homonym or another (20). Where most of us find some ambiguity in language, Malinowski reads the mind of the speaker and avoids confusion. When *pwaypwaya* is used in the sense of *earth* it is regarded as homonymous with *pwaypwaya* used in the sense of *land*. Since all words have some range of meaning, each word represents a group of homonyms, and in this sense is untranslatable by a single equivalent. The translation of untranslatable words, that is, of all words, is accomplished partly by mentalistic mind reading, partly by knowing the context.

In Trobriand, we can divide words into parts of speech: noun, verb, pronoun, adjective, preposition, adverb, and conjunction stand out as clearly as in English by the simple device of translating a word by an English noun when the root is nominal and by a verb when it is an action word, and similarly with adverbs, adjectives and so on (30, 31). This device is indeed simple but it begs the question. Having learned to recognize a part of speech by its formal features and its distribution, it is possible to return to a generalization of meaning of a class of words sharing the same features and distribution. By beginning with meaning Malinowski arrives at the curious conclusion that the parts of speech of Trobriand are precisely the same as those of English. We conclude, on the other hand, that Malinowski has failed to make a linguistic analysis of Trobriand.

In all communities, certain words are accepted as potentially creative acts: you utter a vow or you forge a signature and you may find yourself bound for life to a monastery, a woman or a prison; in each case words are equally powerful causes of action (53). For the Trobriand Islanders, contracts are not only verbal but are effected by non-verbal ritual, as the giving and accepting of certain objects of value. Gifts are neither given nor received without "the phraseology of gifts" (54). By virtue of describing both the act of giving and the phraseology of gifts, Malinowski's treatment begins to be ethno-linguistic. From the phraseology of gifts to the phraseology of ritual is but a short step for Malinowski; yet there is a difference not to be overlooked. In contractual relationships two or more individuals are involved; in religion on the Trobriand Islands as elsewhere, the person or power addressed is apt not to be present, so that all technical descriptions of the phraseology of ritual, including Malinowski's, resolve themselves essentially into narrative texts.

If we wanted to emphasize the opposition between words when they are "idle" and words when they are a matter of life and death, we could take any situation where words mean life or death to a human being; an imperative, a noun, an adjective, even an adverb, screamed from a distance in the dark might reorientate completely the movements of the rescuers or those in danger. The crux of the problem is how to analyze such life and death situations. We may say: (1) our hero called for help;

pregnancy), may later be shown to be stimulated by cultural expectation. Accordingly, neither the historian treating of past cultures, nor the anthropologist dealing with present cultures is ever half as comfortable as is the linguist in excluding any datum as irrelevant.

The elements in linguistics may be stated exhaustively: given an utterance from a language, it is possible to identify that utterance completely in terms of its constituent phonemes and morphemes. Compare with this a segment of behavior from a given culture, as for example leaching acorns among the Tübatulabal. It is easier to ask than to answer what questions about this process shall be included in our field investigation, what questions shall be excluded: shall we give the age and sex of the person engaged in leaching; shall we note the time of day when leaching takes place; shall we clock the operation and come to a statistical average for the time involved in leaching; or shall we merely (or, and also) record what ideal or proper time is said by informants to be required to leach a certain quantity of acorns?

Now compare with these random, somewhat arbitrary questions, the kind of questions which specialists in material culture actually ask about acorn leaching in California, and note that such specialists show considerable agreement in asking such questions (although the questions do not pretend to arrive at an exhaustive statement of the process): whether the acorns are leached with hot or cold water, how many times water is applied to the acorns, whether in a sand basin or in a dirt basin, whether the water is dripped through pine sprigs or through other material, and so on.

Professional questions such as these are based on an awareness of comparative, that is, more or less neighboring cultures; without this awareness, the initial quest for culture elements remains either arbitrary or without boundary; the ethnographer asks whether the Tübatulabal drip water through pine sprigs because other tribes in California use this material (or some other) for the same purpose.

The elements in anthropology (culture traits) are obtained by a sophisticated awareness of their comparative implications; and it does not matter in

or (2) we may quote the hero's very words; and (3) we may cite not only the precise words used but also observe, with pedantic delight, that our hero employed "even an adverb." The problem is at bottom one of correlation: if non-verbal behavior is different (or the same) in emotional crises, is the linguistic structure in the accompanying utterance affected?

To divide anthropology, as one of the leaders of our science has done recently, into three disciplines, one of which is concerned with the human frame, the other with culture, the third with language—shows that the relation between language and culture has been insufficiently appreciated (vii). Everyone who has succeeded in making a competent linguistic statement, including Franz Boas, has treated linguistics as a separate discipline, with techniques of its own. The relationship between a language, once analyzed, and a culture, once described, may be appreciated (or neglected) thereafter.

basic definition whether these traits are obtained in original field-work, as by Kroeber and his colleagues at California, or from the published literature, as by Murdock and his colleagues at Yale. But it does seem to matter, in the sense of affecting the basic definition, when one or another school of anthropology is either unaware of or uninterested in comparative implications; as a structural linguist sticks to one language in a given study, so a functional anthropologist, for example, sticks to one culture, and with a parallel objective, because he too attempts to give the distribution or, as he calls it, the function of each trait in that culture. But he cannot treat the distribution (or function) of culture traits until he has a method for obtaining culture traits in the first instance, and he cannot find boundaries between culture traits and culture complexes unaided by comparative considerations.

Cultural anthropology is dependent upon comparative considerations for finding its elements; linguistics is not. Linguistic analysis provides an exhaustive list of its elements (thus, there are between a dozen and a score or two of phonemes for any given language); cultural analysis does not.

Having found all the elements in the structure he is studying, the linguist explains them (i.e., makes statements about sets of phonemes and sets of morphemes) by giving their distribution relative to each other within the utterances of a single language. The ethnologist looks for his elements or groups of elements (without sharp distinction between one trait and a complex of traits) with a knowledge of parallel occurrences among neighboring cultures, and having thus found his culture traits the ethnologist may (or may not) relate them in an internal economy within the single culture he is primarily investigating.

Use of native language in the study of culture constitutes an associated observation rather than a tool of ethnology.

A few years ago some half dozen ethnologists published papers in the *American Anthropologist*⁵ on the value of linguistics as a tool in ethnology. Stated quantitatively: given a field-trip of a limited period of time, as the tool-use of language is increased, the time available for non-verbal investigation is lessened; what is an efficient proportion of time for each? One can have no quarrel with the tool-use of language, whatever preference is expressed in time devoted to gaining some sort of communicating command of the language. When so used, this kind of knowledge of the language may make it easier to get ethnographic data, but it probably does not add anything to the ethnography which would not be equally available under conditions which permitted the ethnologist to work with completely bilingual informants.

For additional insights into the culture, tool-use of language is, strictly

⁵ Elkin, 1941; Henry, 1940; Lowie, 1940; Mead, 1939.

speaking, irrelevant; for insights beyond what a bilingual informant is able to convey, the ethnologist needs (1) to recognize the linguistic structure of utterances which are part of ethno-linguistic situations; (2) to make detailed observations on how the linguistic utterance relates to the rest of the cultural situation. If an ethnologist were observing features connected with hospitality among a tribe which spoke a language like German, with a distinction between formal *Sie* and informal *du* for "You", he would have to learn that the imperative *Pass auf* ("Look out!") is structurally equivalent to *du* in other utterances, (1). He might correlate the use of utterances characterized by *du* structures with certain kinship relationships, and *Sie* structures with formal guest relationships; then if a guest who was in the formal category used the *du* structure rather than the presumably appropriate *Sie* structure, the ethnographer would be immediately prepared to observe how the host reacted to such behavior contrary to cultural expectation, (2).

Problems which linguistics has in common with other fields are only partly shared by cultural anthropology.

Throughout the formulation of linguistic structure there are problems of a mathematical or logical nature; in determining the physical bases of phonemes and the manner in which phonemes are produced by speakers, linguists turn to physics and physiology. Problems of the type here mentioned do not occur in cultural anthropology which has, however, other points of contact with these fields, as in the study of diet (physiology).

There remain a number of fields in which linguistics and cultural anthropology share many contacts, but with different emphasis or degrees of interest: semantics, theory of signs, philosophy; psychology, psychiatry, psychoanalysis; oral literature, written literature. Stylistic studies are often recommended but rarely attempted. There are also fields in which cultural anthropology has wide contact and linguistics restricted contact, such as geography.

2. TRENDS IN LINGUISTICS

A central interest in modern linguistics is the synchronic description of one language at a time.

In descriptive linguistics, a form is regarded as explained when its place is found in the total structure of a single language. In historical linguistics, a form is regarded as explained when the forms from which it developed are reconstructed, or found in early records. During the nineteenth century, formulations of the Indo-European comparative method were brought to full expression.

In contrast to this developmental work on comparative formulations in the nineteenth century, much of the work of the present century has been devoted to formulating descriptive statements about diverse languages (without respect

to the historical importance of the speakers of such languages, or to whether written records were or were not available). This interest, in its characteristically comprehensive expression, was very demanding, the more so because the languages of the world are very numerous and very diversified. It turned out that it was not often profitable in following the dominating descriptive interest also to explain or state linguistic forms in one of these languages relative to earlier forms, or to forms in some other language. This is all that is generally meant by the statement that linguistic interest has shifted from historical to descriptive work.

The fact that linguistics permits exact statements has led to experimentation in compact and highly organized descriptions.

The older grammars showed some interrelations among the materials treated, usually in the form of paradigms; otherwise the materials were presented with something of an archival organization (with all materials on the nouns in one chapter, materials on the adjectives in another),—perhaps convenient but not compact. Such organizations often followed categories of meanings which reflected some kind of translation from Latin or other western grammatical tradition. These grammars were often neither exact nor consistent. For unwritten languages, a given word or morpheme (if not normalized) would be recorded in several different ways. The first great advance in the direction of exactness and consistency came with phonemic writing.

The old grammars listed facts about a language and (often sporadically) the attendant conditions. Recent grammars tend to set up the elements of a language in such a way that the attendant conditions are included in the definition of the elements. The old grammars listed exceptions to their rules. Recent grammars tend to state asymmetrical features in the same fashion as symmetrical features of the language being described; then no methodological distinction results between rules (symmetries) and exceptions (asymmetries).

Applied linguistics is used in education and in social control.

We use the term “applied linguistics” for the entry of research linguists into the field of practical teaching, dictionary work, code work, devising of alphabets, and administrative problems concerned with subject populations. The part that descriptive linguists play in such work is very small, but it has already had some effect in the fields here mentioned.

In language teaching in particular the chief effects include: (1) the use of a phonemic orthography or at least some sophistication in phonemics; (2) in grammatical presentation, the use of formal criteria as distinguished from meaning considerations; (3) partial use of the linguistic structure of a given language, namely, the isolation of small areas in the morphology and the arrangement of these, with illustrative models, for learning the language.

Structural comparability of languages may be stated independently of their genetic relationships.

By the side of the major comparative and historical work of nineteenth and twentieth century linguistics, it is possible to distinguish a few approaches to the problem of linguistic types, including an approach which is just beginning to be formulated: (1) grammatical features of Indo-European served as a basis of reference; (2) exotic features of aboriginal languages were noted and these raised the problem of linguistic diversity; (3) grammars were compared more or less implicitly by juxtaposing brief summaries of related or unrelated languages between the covers of one book; (4) early attempts at generalizing were based on arbitrarily selected key characteristics; premature classifications served to stimulate a controversy on genetic relationship versus borrowing which led away from the problem of linguistic types; (5) structural comparability may lead to a solution of the problem of linguistic types. The discussion paragraphs that follow are numbered to correspond with these various approaches.

(1) From the first explorations in comparative work, similarities and differences among genetically related languages could hardly escape attention. Linguists recognized not only cognate words, but also parallel constructions and paradigms. Features which were recurrent among Indo-European languages focussed attention in two essentially different ways: if any language not belonging to the Indo-European family showed some features characteristic of Indo-European, these received central attention; if a language belonging to the Indo-European family failed to have one of the features otherwise recurrent in the family, that was emphasized.

(2) As more languages became known, features were noted even though they might not occur in Indo-European. Thus, for Melanesian languages, the luxuriance in the expression of number was frequently cited: speakers distinguish between *I* (singular), *we two including you* (dual inclusive), *we two including him but not you* (dual exclusive), *we three* (trialis), and *we all* (plural). For Bantu languages of Africa it was discovered that there were often several classes of nouns which marked such notions as person class, wooden-object class, and so on. For American Indian languages, which gave the greatest impetus to the recognition of linguistic diversity, investigators found such unexpected forms as those marking fourth person (the less important or conspicuous of two third persons; in *the man and his wife*, for example, *the man* is marked as a third person, *his wife* as a fourth person), and such noun classes as animate and inanimate, or long object versus round object. Sometimes such features were presented as though they had replaced somewhat similar features in Indo-European or Semitic; instead of sex gender, Algonquian marks animate-inanimate gender, and so on.

(3) It is clear that those who presented such features as these were inter-

ested in comparing grammars. Their method of presentation was to juxtapose summaries of the languages treated, brief paraphrases of the full-length grammars, with special attention given to particular word classes, such as personal pronouns. Juxtaposed summaries of grammars were made both for unrelated languages (as in Finck's⁶ famous book on language types), and for the branches of a single language family (as Bergsträsser's⁷ work on Semitic).

(4) The early attempts to generalize these comparisons were satisfied by selecting, rather casually, key characteristics in respect to which various languages differed or were similar. Languages like Turkish were called agglutinative, Latin and the like were called inflectional, Chinese (and sometimes English) was called isolating, and most native American languages polysynthetic. In some cases languages which showed similarity of this order were grouped together with the implication that a genetic relationship existed which was capable of later demonstration; opponents of this implication suggested an alternative explanation which is often discussed as part of the larger subject of areal linguistics. Besides the implication of genetic relationship (that is, historical connection) among languages of the same generalized type (as proposed by Sapir for native America), there were also theories which stated that one type developed from the other by an evolutionary process or progress (more especially for Old World languages).

(5) The current view on this matter is that implications of any sort, whether of a genetic or diffusional character, will necessarily be delayed until the structure of each language is stated in a way to admit of controlled rather than casual comparability. Having first stated comparable structures, it may then be possible to discover language types so that any particular language can be associated with one type or another.

Historical and comparative techniques are insufficiently applied to aboriginal languages.

When nothing more than fragmentary word lists were available for aboriginal languages, it was customary to make preliminary estimates of genetic relationship based on some regularity of sound correspondences, but usually not on reconstructions of the parent languages for each family. Such reconstructions are still largely lacking. Workers in aboriginal languages have not contributed greatly to what is, nevertheless, the great triumph of linguists, namely, the comparative method.

This method deals with the reconstruction of linguistic forms no longer spoken as the parents of later cognate forms used (for the most part) in genetically related languages. Much of Proto-Indo-European was reconstructed in this way; Old French reconstructions were so exact that in some cases they

⁶ Finck, 1923.

⁷ Bergsträsser, 1928.

yielded forms which were lacking in Latin but which were subsequently found in late Latin manuscripts. Among American Indian languages the most important example of the application of the comparative method is the reconstruction of Central Algonquian, chiefly by Bloomfield.

By the side of the comparative method there are other techniques for historical attestation and inference, resting largely on the evidence of written records. Such techniques make it possible to say, for example, that certain German loanwords in Lithuanian were borrowed at an earlier period than certain others. (German words which shared the effects of certain historically known Lithuanian changes must have been borrowed by Lithuanians before those changes occurred; and German loanwords which would have been subject to these changes but did not show them must have entered Lithuanian after the changes had ceased.) Such techniques can be in part applied even when written records are lacking. Thus, Algonquian students first reconstruct a parent language, PA, and then find that certain daughter languages share certain phonemes: (1) PA θ , l , n are kept distinct in Cree; (2) PA θ and l coincide as l , but l is kept distinct from n in Shawnee, Miami-Peoria, Delaware, Penobscot and other eastern Algonquian languages; PA θ and l and n coincide as n in the remaining Central Algonquian languages. In respect to the three phonemes here noted for the parent language, Cree is most archaic; Cree did not join in an innovation affecting some Central and some Eastern languages which must have been contiguous at the time of this innovation; only a few Central languages followed the final innovation.

Increased experience with linguistic structure, which characterizes twentieth century linguistics, has led investigators in both Europe and America to think about the use of the comparative method and historical analysis on whole genetically related language structures rather than merely on groups of cognate words. For American Indian languages, where historical research has been meager though successful, the controlled reconstruction of both groups of words and language structure will no doubt yield much information.

Dialect geography and diffusional areal studies offer a new approach to historical problems in anthropology.

Dialect geography is a specialized branch of linguistics, growing by the side of historical linguistics (which employs the comparative method), and descriptive linguistics. Most dialect geography surveys have mapped distributions of isolated grammatical features, pronunciation of particular vowels (and consonants), and the use of different words for the same item in material culture; loanwords are sometimes separately mapped. The primary interest in dialect geography is the study of linguistic diffusion among people who speak a single language (in the sense that the speech of neighbors is mutually intelligible).

In maps which show the distribution of, say, the various pronunciations of the word for "make" in German, forms with *x* (*maxen* = machen) are found south of an east-west axis; north of this axis, forms with *k* (*maken* = machen) are found. The east-west line separating the two areas is called the *x-k* isogloss.

Besides isoglosses that divide the whole area in which a language is spoken into roughly two parts, as in the example above, there are cases in which a particular word or pronunciation occurs in one or more restricted, separated regions within the whole area. The lines enclosing such sub-areas or insular regions are also called isoglosses.

As multiple features are mapped, some will share the identical distribution of others or overlap in some regions of the whole area. When, as frequently happens, the isoglosses for different features do run parallel to each other or enclose roughly the same insular areas, we speak of bundles of isoglosses.

A bundle of isoglosses, reflecting parallel distribution of several linguistic features, may extend halfway across the area in which a given language is spoken, and then fan out at that part of the area in which the overlapping distribution of features is interrupted by eccentric or other kinds of independent distributions which cannot be represented by major bundles of isoglosses. Major as well as minor isoglosses may cross each other.

Statements on areal interpretation are elaborate or simple, depending on whether the isoglosses show an elaborate distribution in bundles with fanwise dispersal and crisscrossing of individual isoglosses, or a simple distribution of bundles of isoglosses which remain neatly together as bundles over or within the entire area. The total result of formulations by means of isoglosses leads to a somewhat different kind of mapping than that made for culture areas in anthropology, but the diffusional interpretation of both is much the same.

The historical interpretations made on the basis of the major European and American dialect geographies have been corroborated by external data on movements of population, lines of trade, and changes of political boundaries. In places where there are few external data of this nature available, and where archaeological horizons do not integrate with the culture of the people being studied linguistically, dialect geography may provide the only source of historical interpretation.

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